

NEKHEL, Egypt

WMO: 624520

Lat: 29.9142N

Lon: 33.7402E

Elev: 403

StdP: 96.58

Time Zone: 2.00 (E02)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.1	1.1	-10.2	1.7	11.4	-7.7	2.1	12.9	15.0	9.0	12.9	11.5	2.8	250	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	37.9	18.5	36.6	18.3	35.4	18.3	22.7	27.4	22.1	27.1	21.6	27.0	4.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	17.2	24.1	20.9	16.3	23.8	20.1	15.5	23.7	69.3	27.2	66.9	27.2	64.8	26.7	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.7	9.4	8.2	DB	-2.6	40.9	2.1	1.4	-4.1	41.9	-5.4	42.7	-6.5	43.5	-8.0	44.5	
			WB	-4.3	25.0	1.7	2.1	-5.6	26.5	-6.6	27.8	-7.6	29.0	-8.9	30.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.1	9.8	11.9	14.8	18.3	22.3	25.1	27.0	27.1	24.8	21.4	15.7	11.2
	DBStd	6.63	2.54	3.31	3.56	3.44	3.09	2.12	1.52	1.41	1.95	2.66	3.13	2.45
	HDD10.0	64	34	14	2	0	0	0	0	0	0	0	1	14
	HDD18.3	933	265	183	123	42	3	0	0	0	0	5	91	220
	CDD10.0	3402	27	66	149	249	380	452	526	529	446	352	173	53
	CDD18.3	1229	0	2	13	42	125	202	268	271	196	98	13	0
	CDH23.3	15574	9	59	264	771	1778	2701	3541	3316	2077	906	145	8
CDH26.7	8046	1	14	97	351	876	1497	2063	1865	968	284	31	1	
Wind	WSAvg	3.3	3.3	3.6	3.8	3.8	3.5	3.6	3.2	3.2	3.2	3.1	2.9	2.9
Precipitation	PrecAvg	29	7	4	3	1	1	0	0	0	0	3	3	5
	PrecMax	62	20	17	24	7	7	0	1	0	2	12	26	20
	PrecMin	11	1	0	0	0	0	0	0	0	0	0	0	1
	PrecStd	13	5	3	5	2	2	0	0	0	0	3	6	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.2	28.7	33.6	37.0	39.1	40.0	39.5	38.6	36.4	34.6	30.1	24.6
		MCWB	11.8	14.5	15.2	16.3	17.3	17.9	18.8	19.7	18.7	17.7	16.7	12.1
	2%	DB	20.9	24.9	29.3	33.6	36.2	37.1	37.5	36.8	34.5	31.6	27.1	21.2
		MCWB	10.2	12.1	13.9	14.9	16.8	18.1	18.8	19.3	19.2	18.3	15.7	11.3
	5%	DB	18.3	21.8	26.4	30.9	33.9	35.4	36.3	35.6	33.1	29.6	24.8	19.4
		MCWB	10.1	11.2	12.9	14.3	16.4	17.9	18.6	19.0	19.4	18.3	15.0	11.3
	10%	DB	16.4	19.2	23.1	27.8	31.7	33.9	35.1	34.5	31.7	27.9	22.8	17.9
		MCWB	9.6	10.4	11.9	13.6	16.0	17.8	18.6	19.1	19.4	18.1	14.5	11.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.2	15.6	16.4	17.4	19.2	21.6	23.3	23.7	24.1	22.1	19.3	14.5
		MCDB	19.9	24.5	27.9	31.3	28.5	26.4	27.5	28.4	28.1	27.4	23.7	19.2
	2%	WB	12.0	13.5	15.0	16.2	18.2	20.6	22.1	22.7	22.5	20.8	17.9	13.3
		MCDB	18.2	20.7	25.3	29.3	29.0	27.3	27.2	27.3	27.5	25.5	22.3	18.2
	5%	WB	10.9	12.2	13.7	15.2	17.5	19.8	21.5	22.2	21.7	20.1	16.9	12.4
		MCDB	16.8	19.2	23.9	27.2	28.9	28.0	26.9	27.3	26.6	24.9	22.1	17.9
	10%	WB	10.0	11.3	12.6	14.4	16.9	19.2	20.8	21.7	21.1	19.5	15.6	11.4
		MCDB	15.7	17.8	21.3	25.4	28.2	28.2	27.0	27.3	26.5	24.6	21.8	17.0
Mean Daily Temperature Range	5% DB	MDBR	13.8	14.1	15.1	16.6	17.3	17.3	16.5	15.5	14.4	13.7	14.5	14.2
		MCDBR	17.0	18.2	20.0	21.9	20.3	19.9	18.4	17.2	15.9	15.3	16.1	16.4
		MCWBR	9.7	9.1	8.9	8.8	7.2	6.6	5.7	5.3	5.3	5.9	7.6	9.5
	5% WB	MCDBR	14.5	15.0	17.4	18.1	17.7	17.1	15.6	15.1	13.5	12.9	13.2	14.0
		MCWBR	9.0	7.8	7.9	7.8	6.4	5.9	5.0	4.9	4.6	5.0	6.2	8.9
Clear-Sky Solar Irradiance	taub		0.346	0.381	0.432	0.481	0.488	0.408	0.408	0.408	0.415	0.423	0.361	0.340
	taud		2.209	2.089	1.924	1.813	1.824	2.075	2.093	2.100	2.080	2.044	2.207	2.249
	Ebn at Noon		873	872	849	820	815	878	876	874	855	816	848	861
	Edh at Noon		119	147	185	215	214	165	162	159	157	152	118	109
All-Sky Solar Radiation	RadAvg		3.68	4.59	5.92	6.99	7.74	8.39	8.13	7.52	6.52	5.20	4.09	3.45
	RadStd		0.14	0.16	0.22	0.17	0.22	0.08	0.12	0.10	0.13	0.16	0.11	0.12

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	+0.40	N/A	-1.06	N/A	N/A	N/A	N/A	N/A	+135	+146

Nomenclature: See separate page